



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Objective**

- Predict Falcon 9 first-stage landing success using historical launch data and machine learning.

- **Summary of Methodology**

- Data Collection through **SpaceX REST API**.
 - Data Collection using **Wikipedia Web Scraping**.
 - Data Wrangling and Feature Engineering.
 - Exploratory Data Analysis (EDA) using SQL Queries, Python Visualizations (Matplotlib & Seaborn), and Plotly Dash.
 - Interactive Visual Analytics using **Folium** and **Plotly Dash**.
 - Predictive Analysis using **Classification Models**.

- **Summary of Results**

- Landing success rate improved significantly over time.
 - Payload mass, orbit type, and launch site influenced landing success.
 - SQL and visualizations revealed important launch patterns.
 - Interactive dashboards provided deeper insights into launch data.
 - The best classification model achieved the highest prediction accuracy.

Introduction

- **Project Background and Context:**
- SpaceX has significantly reduced the cost of space missions by successfully recovering and reusing Falcon 9 first-stage boosters. Traditional rocket launches are expensive because rockets are usually discarded after launch. By predicting whether the Falcon 9 first stage will land successfully, organizations can estimate mission costs, improve planning, and support data-driven decision-making. This project uses historical SpaceX launch data to analyze landing outcomes and build machine learning models for landing prediction.
- **Project Objective:**
- The project aims to identify the factors influencing Falcon 9 first-stage landing success and develop a machine learning model to predict landing outcomes.
- Collect Falcon 9 launch data using the SpaceX REST API and **Wikipedia Web Scraping**.
- Perform **data wrangling, SQL analysis, and exploratory data analysis (EDA)**.
- Build interactive visualizations using SQL, Folium, and Plotly Dash.
- Develop, evaluate, and compare machine learning classification models to predict Falcon 9 first-stage landing success.

Section 1

Methodology

Methodology

Executive Summary

- **Data Collection Methodology**

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Collected additional launch records through **Wikipedia Web Scraping**.

- **Data Wrangling**

- Cleaned missing values and standardized the dataset.
- Created the landing success label and prepared features for analysis.

- **Exploratory Data Analysis (EDA)**

- Performed **SQL queries** to analyze launch statistics.
- Created visualizations using Matplotlib and Seaborn.

- **Interactive Visual Analytics**

- Built an interactive **Folium** map to analyze launch site locations.
- Developed an interactive Plotly Dash dashboard with dropdown filters, charts, and data visualizations for launch analysis.

- **Predictive Analysis**

- Built, trained, and evaluated multiple machine learning classification models.
- Compared model performance and selected the best-performing model based on prediction accuracy.

Data Collection

- **The data was collected using various methods**
- Data collection was done using get request to the SpaceX API.
- Next, we decoded the response content as a Json using `.json()` function call and turn it into a pandas dataframe using `.json_normalize()`.
- We then cleaned the data, checked for missing values and fill in missing values where necessary.
- In addition, we performed web scraping from Wikipedia for Falcon 9 launch records with BeautifulSoup.
- The objective was to extract the launch records as HTML table, parse the table and convert it to a pandas dataframe for future analysis.

Data Collection - SpaceX API

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Retrieved launch date, payload mass, orbit, launch site, booster version, and mission outcome.
- Converted JSON responses into a structured Pandas DataFrame for analysis.
- **Right side:** SpaceX API notebook screenshot.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

```
import requests
import pandas as pd

static_json_url = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json"

response = requests.get(static_json_url)

print(response.status_code)
```

200

We should see that the request was successful with the 200 status response code

```
response.status_code
```

200

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
# Use json_normalize method to convert the json result into a dataframe
data = pd.json_normalize(response.json())
```

Using the dataframe `data` print the first 5 rows

```
# Get the head of the dataframe
data.head(5)
```

CSV

Data Collection - Scraping

- Extracted additional Falcon 9 launch records from **Wikipedia** using **BeautifulSoup**.
- Parsed HTML tables and converted them into a Pandas DataFrame.
- Used the scraped data to enrich and validate the dataset.
- Successfully extracted Falcon 9 launch tables using BeautifulSoup for further analysis.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

```
# use requests.get() method with the provided static_url and headers
# assign the response to a object
response = requests.get(static_url, headers=headers)
```

Create a BeautifulSoup object from the HTML response

```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content

soup = BeautifulSoup(response.text, "html.parser")
```

```
print(soup.title.string)
```

List of Falcon 9 and Falcon Heavy launches - Wikipedia

Print the page title to verify if the BeautifulSoup object was created properly

```
# Use soup.title attribute
print(soup.title)
```

<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>

```
# Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`
```

```
html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

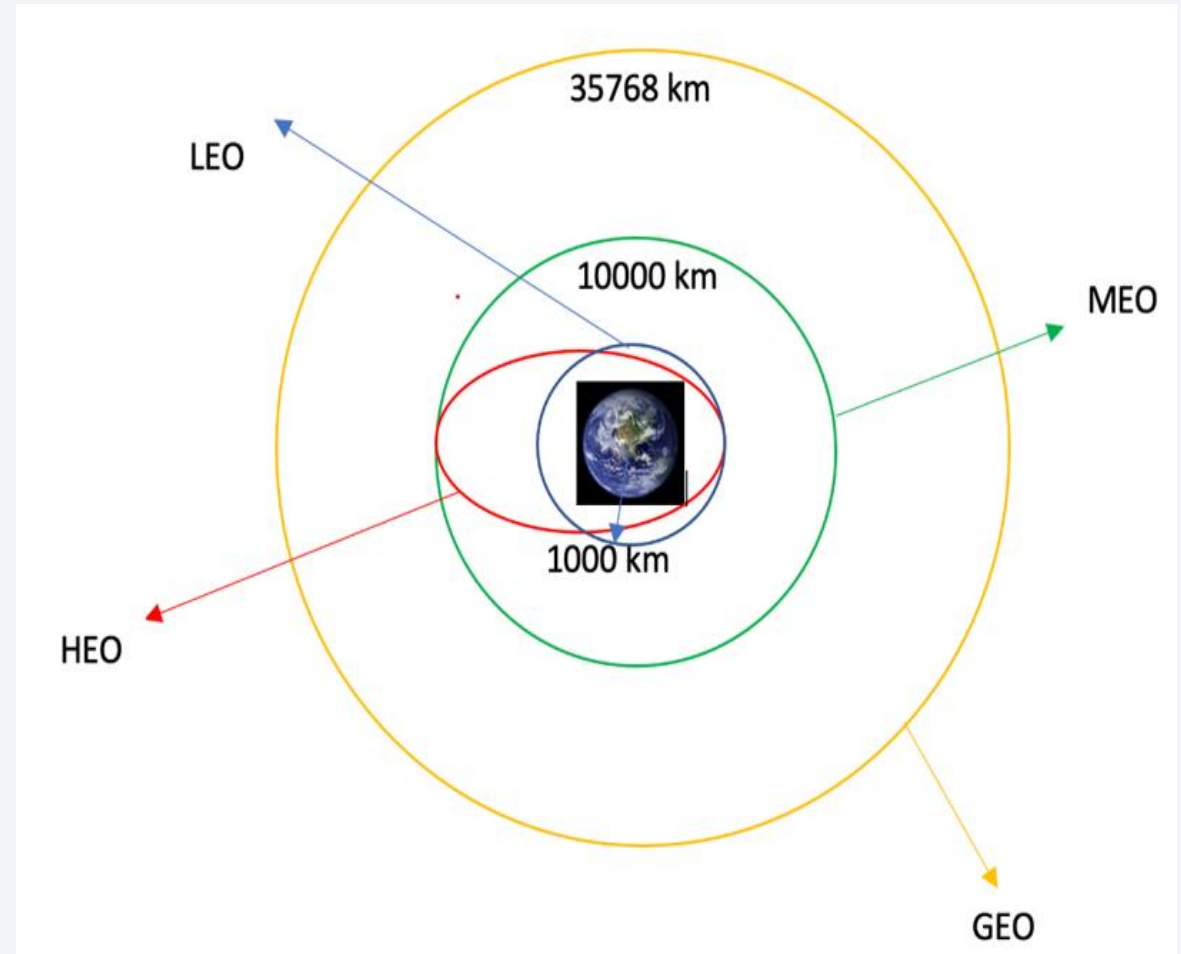
```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
print(first_launch_table)
```

Data Wrangling

- Cleaned missing and inconsistent values.
- Selected relevant features for analysis.
- Created the landing success label.
- Prepared the final dataset for EDA and machine learning.

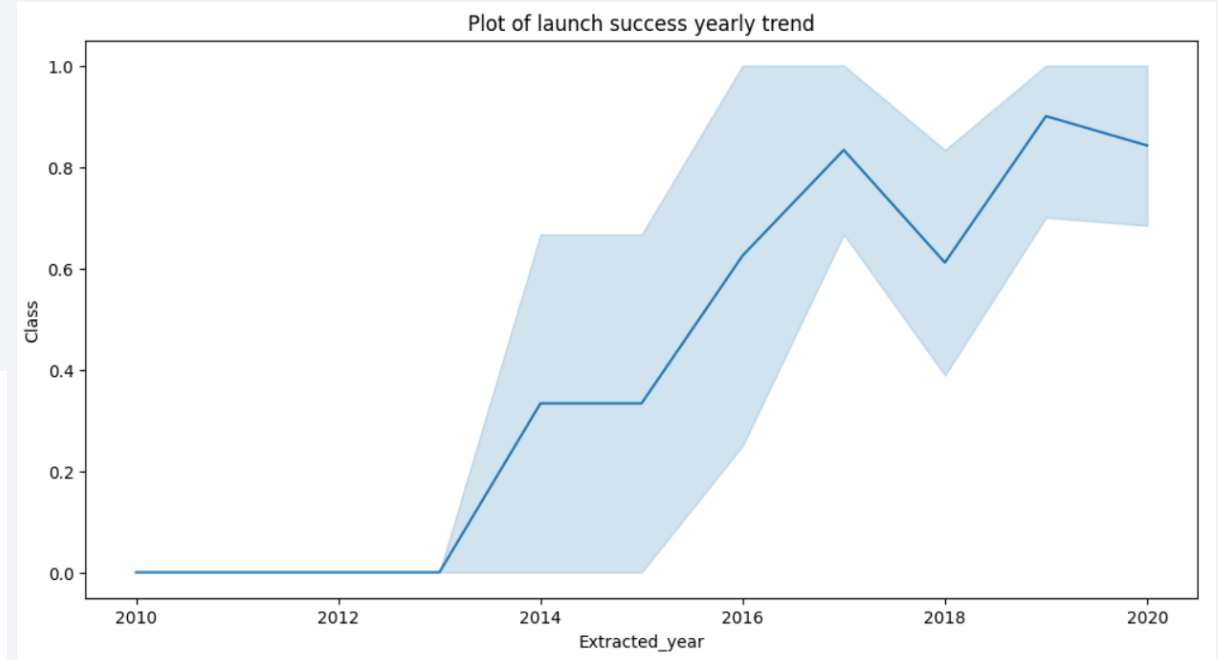
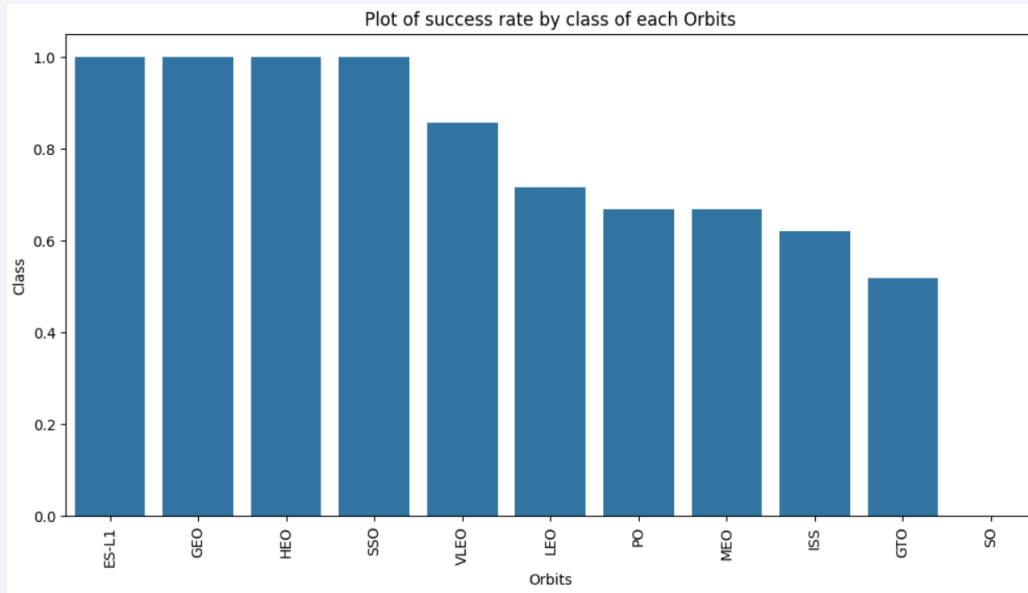
GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>



EDA with Data Visualization

- Launch success increased over time.
- Success rates differed across orbit classes.
- EDA revealed patterns influencing landing success.



GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

EDA with SQL

- Used SQL queries to summarize launch statistics.
- Analyzed launches by launch site, mission outcome, and booster version.
- Identified trends and generated insights from the data.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

Interactive Map with Folium

- Built interactive **Folium** maps to visualize launch site locations.
- Developed a **Plotly Dash** dashboard with filters and interactive charts.
- Enabled dynamic exploration of launch data.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

Build a Dashboard with Plotly Dash

- We built an interactive dashboard with Plotly dash.
- We plotted pie charts showing the total launches by a certain sites.
- We plotted scatter graph showing the relationship with Outcome and Payload Mass (Kg) for the different booster version.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

Predictive Analysis (Classification)

- Built multiple classification models to predict landing success.
- Compared model performance using evaluation metrics.
- Selected the best-performing model based on prediction accuracy.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

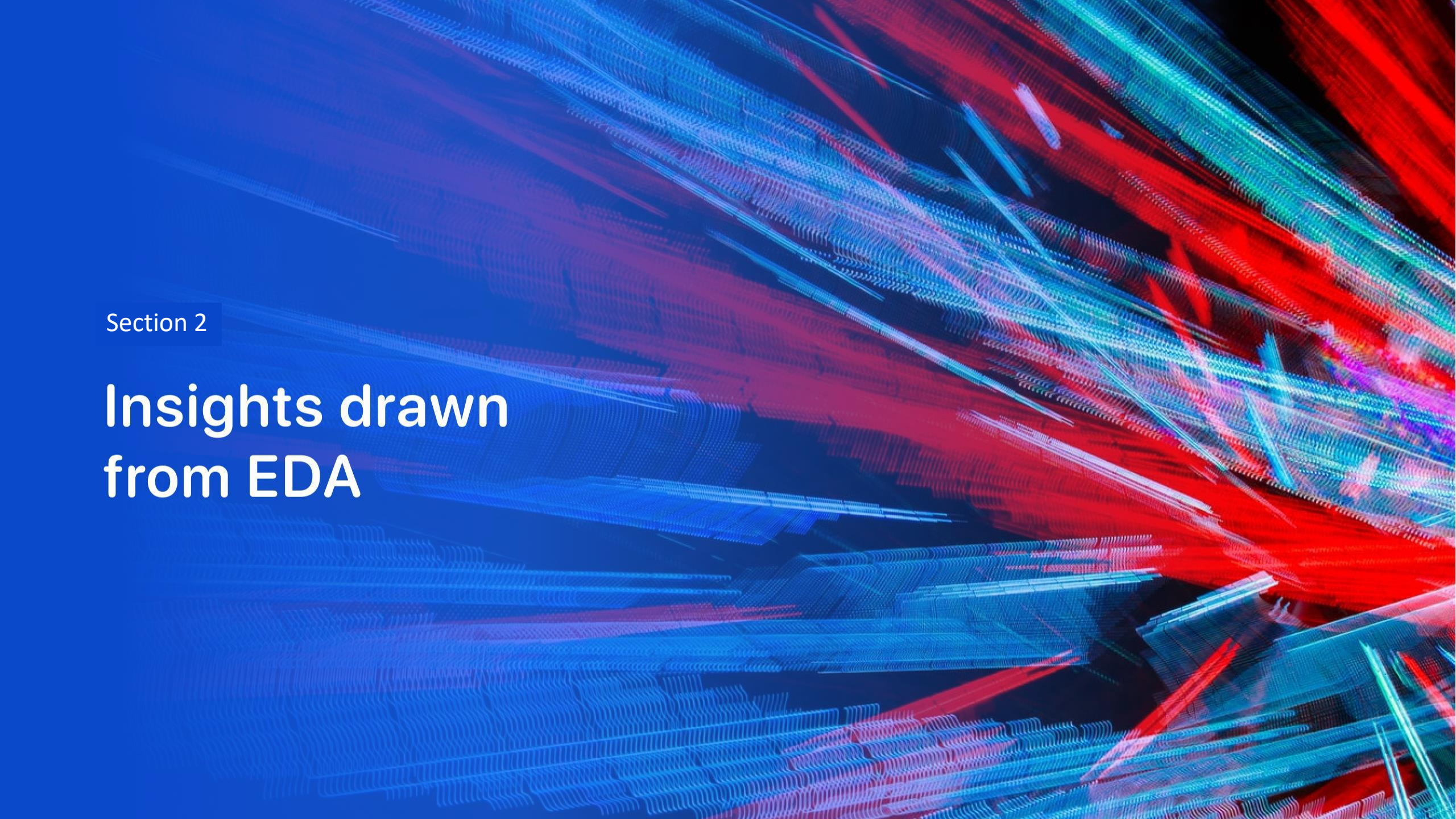
Evaluate Each
Model

Results

- **Landing Success Trend:** This project integrates data collection, EDA, SQL, interactive dashboards, and machine learning to predict Falcon 9 landing success.
- **Key Factors:** Payload mass, launch site, orbit type, flight number, and booster version influenced landing success.
- **EDA, SQL & Visualization:** SQL queries, Folium maps, and the Plotly Dash dashboard provided interactive insights.
- **Predictive Analysis:** The best classification model achieved the highest prediction accuracy.
- **Overall Outcome:** Historical launch data can effectively predict Falcon 9 first-stage landing success and support mission planning.

GitHub Repository:

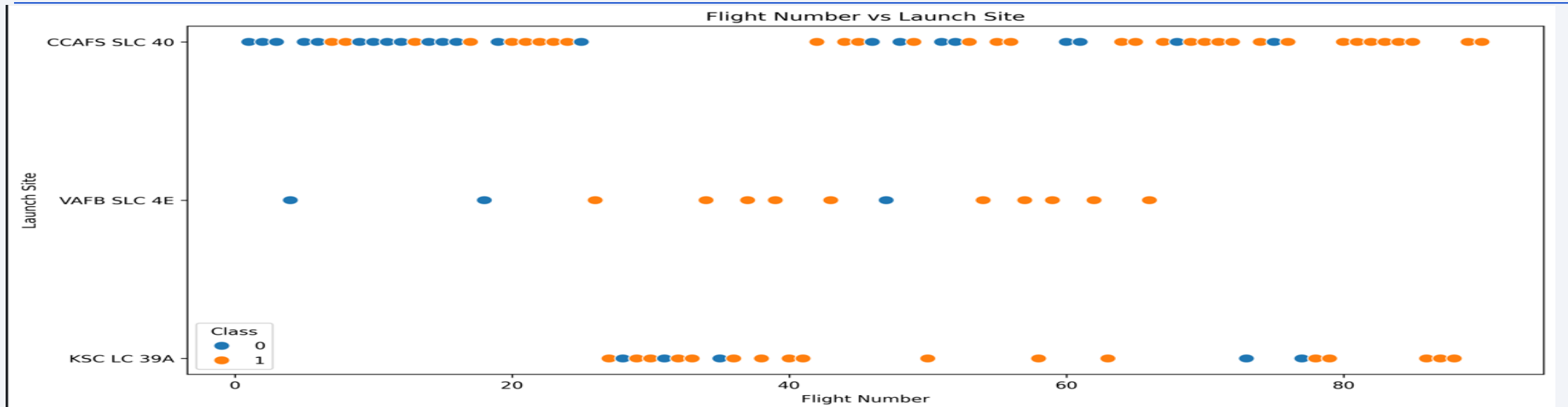
<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. A faint, light blue grid pattern is also visible, particularly in the lower right quadrant, overlaid on the streaks.

Section 2

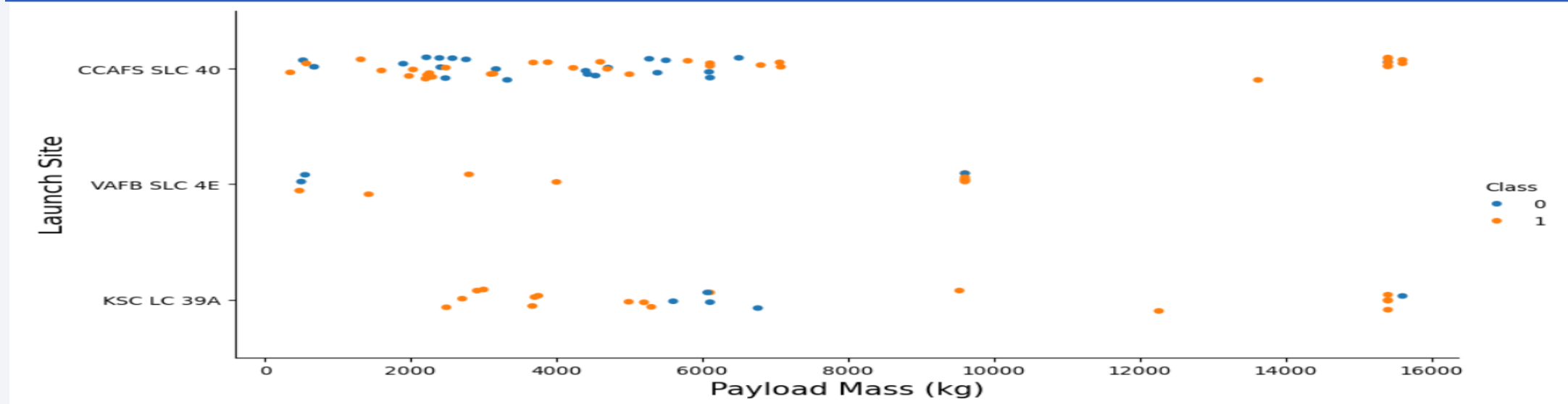
Insights drawn from EDA

Flight Number vs. Launch Site



- Early Falcon 9 missions had lower landing success, while later missions showed higher success rates.
- Launch success improved as the **flight number** increased, indicating continuous improvements in booster recovery and operational experience.
- **CCAFS SLC 40** recorded the highest number of Falcon 9 launches, making it the most frequently used launch site.
- **KSC LC 39A** and **VAFB SLC 4E** also demonstrated successful landings, although with fewer launches than CCAFS SLC 40.
- Overall, both launch site and flight number influenced Falcon 9 first-stage landing success

Payload vs. Launch Site



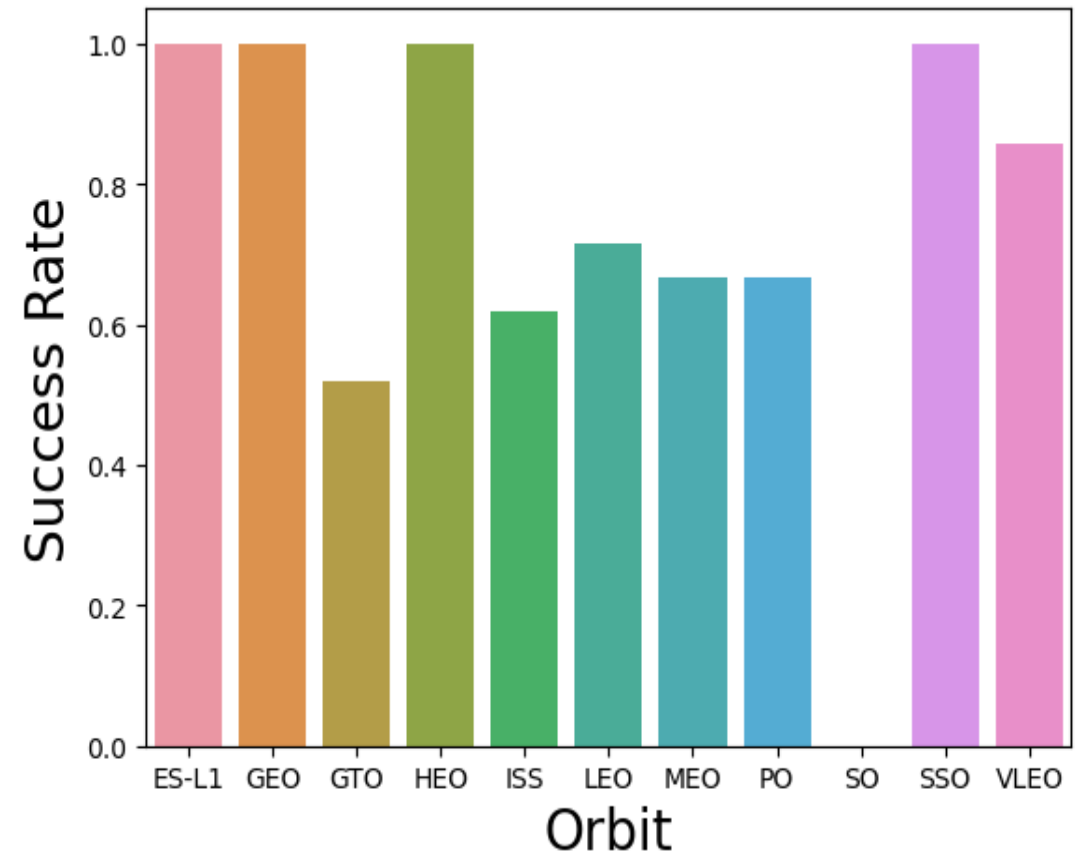
- **CCAFS SLC 40** handled the widest range of payload masses and was the most frequently used launch site.
- **KSC LC 39A** primarily handled medium- to high-payload missions and achieved several successful landings.
- **VAFB SLC 4E** conducted fewer launches but demonstrated successful missions across different payload categories.
- Both **successful** and **unsuccessful** landings occurred at low, medium, and high payload masses, indicating that **payload mass alone was not the primary factor affecting landing success**.
- Overall, launch site and payload mass together influenced Falcon 9 landing success.

Success Rate vs. Orbit Type

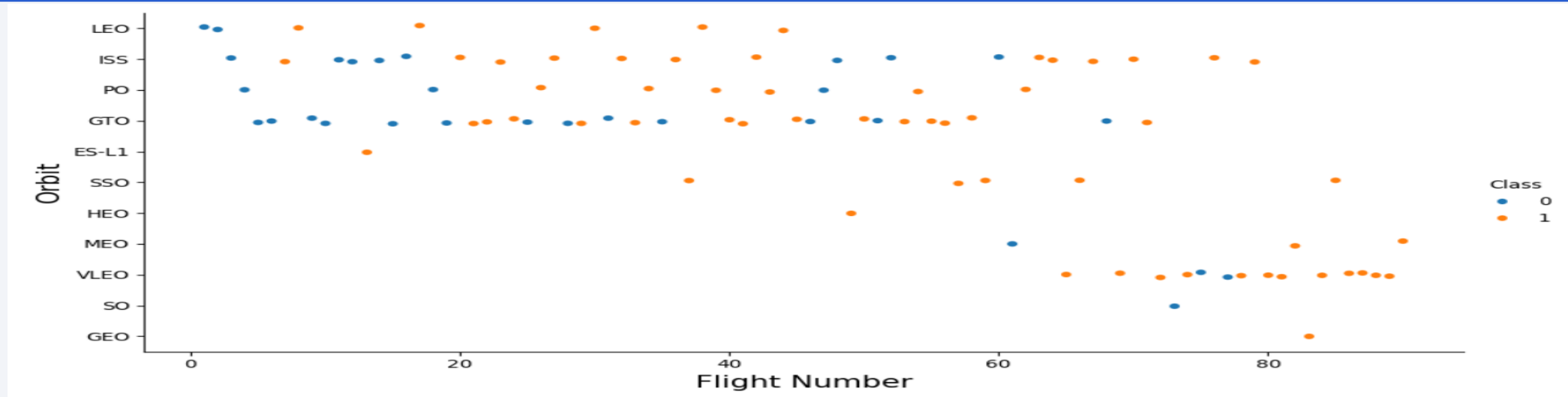
- ES-L1, GEO, HEO, and SSO recorded a 100% landing success rate in the available dataset.
- Success rates varied across orbit types, indicating that orbit type is associated with Falcon 9 landing performance.

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

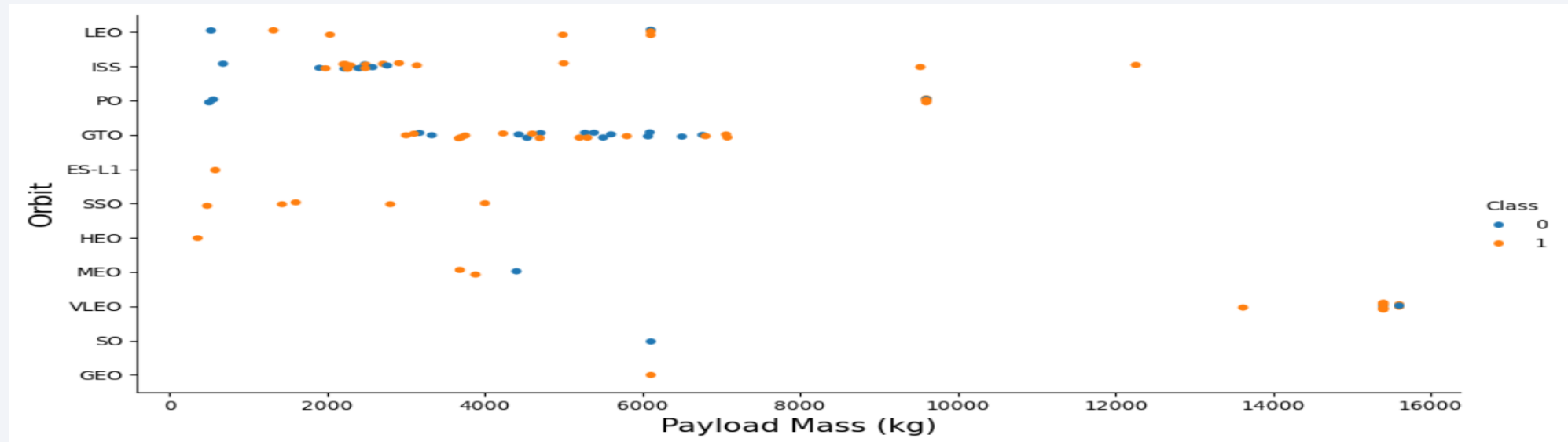


Flight Number vs. Orbit Type



- LEO, ISS, and VLEO missions generally achieved high landing success rates.
- GTO missions showed mixed landing outcomes, with both successful and failed landings.
- Landing success generally improved with increasing flight numbers, especially in later missions.
- Orbit type influenced landing success, while payload mass and launch site also played important roles.

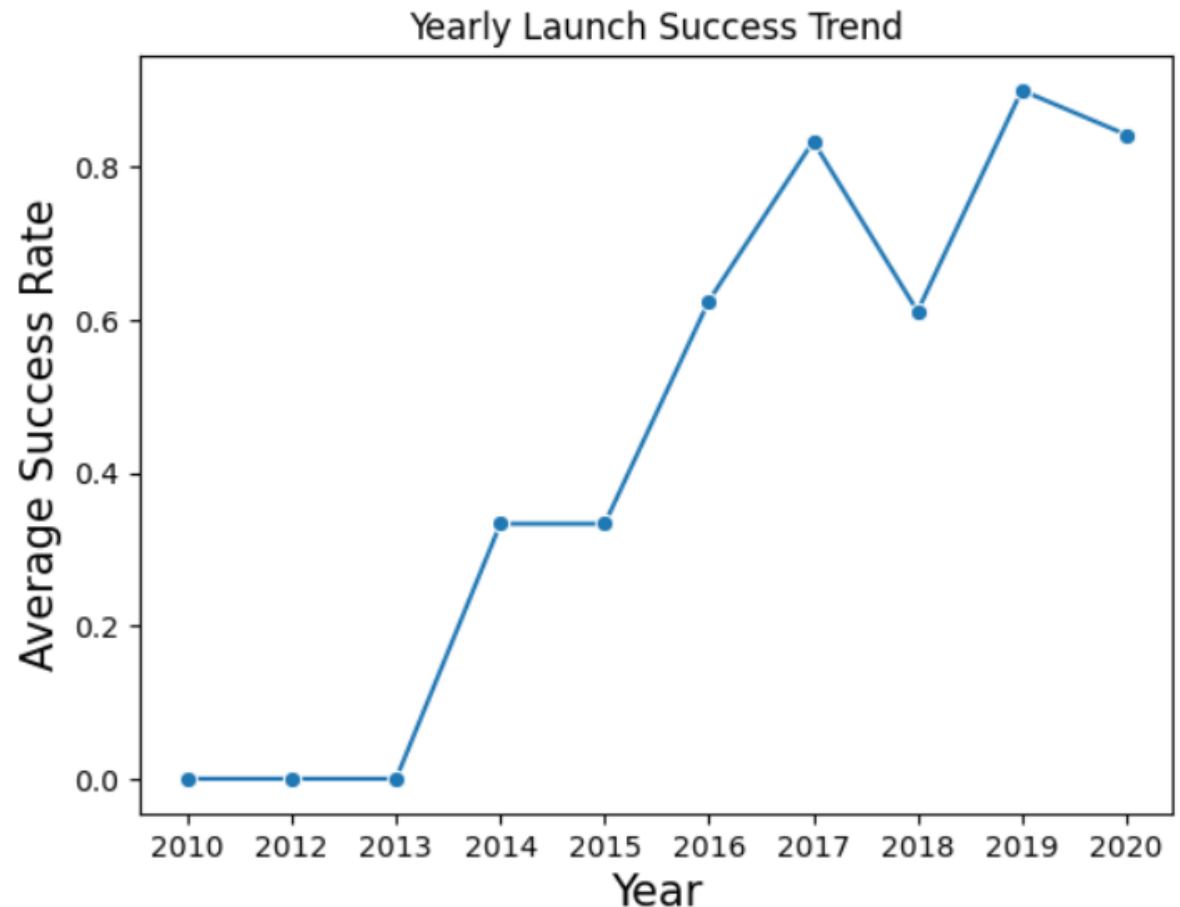
Payload vs. Orbit Type



- GTO missions covered a wide range of payload masses, with both successful and unsuccessful landings observed.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions generally carried low to medium payload masses.
- Payload mass alone did not determine Falcon 9 landing success; orbit type and mission characteristics also influenced landing outcomes.

Launch Success Yearly Trend

- Overall launch success rates improved over the years despite minor fluctuations.
- Success rates were very low during the early years (2010–2013).
- From 2016 onward, average launch success rates increased significantly.
- By 2019–2020, SpaceX consistently achieved high launch success rates, demonstrating improved booster recovery and mission reliability.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
%%sql
```

```
SELECT DISTINCT Launch_Site  
FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

- Identified **four unique SpaceX launch sites** from the dataset.
- Most launches were conducted from **Cape Canaveral** and **Kennedy Space Center**.
- **Vandenberg Space Force Base** was primarily used for polar orbit missions.
- These launch sites were used for further EDA, dashboard development, and predictive analysis.



Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
```

```
SELECT *  
FROM SPACEXTABLE  
WHERE Launch_Site LIKE 'CCA%'  
LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

We used the query above to display 5 records where launch sites begin with 'CCA'

Total Payload Mass

- We calculated the total payload carried by boosters from NASA as 45596 using the query below.

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
```

```
SELECT SUM(Payload_Mass__kg_) AS Total_Payload_Mass  
FROM SPACEXTABLE  
WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Total_Payload_Mass

45596

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[22]: %%sql

SELECT Booster_Version,
       AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1'
GROUP BY Booster_Version;

* sqlite:///my_data1.db
Done.
```

```
[22]: Booster_Version Average_Payload_Mass
      F9 v1.1          2928.4
```

- The Falcon 9 v1.1 booster carried an average payload mass of **2928.4 kg**, which serves as a reference for comparing later booster versions.

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[24]: %%sql
      SELECT MIN(Date) AS First_Successful_Landing_Date
      FROM SPACEXTABLE
      WHERE Landing_Outcome = 'Success (ground pad)';

* sqlite:///my_data1.db
Done.
```

```
[24]: First_Successful_Landing_Date
      2015-12-22
```

- The first successful ground pad landing occurred on **December 22, 2015**.
- This event demonstrated the successful recovery of a Falcon 9 first-stage booster on land.

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

[26]: %%sql

```
SELECT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND Payload_Mass_kg_ > 4000
AND Payload_Mass_kg_ < 6000;
```

```
* sqlite:///my_data1.db
Done.
```

[26]: **Booster_Version**

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

- Four Falcon 9 booster versions achieved successful drone ship landings while carrying payloads between **4,000 kg and 6,000 kg**, indicating reliable recovery performance for medium-payload missions.

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
[55]: import pandas as pd

task_7a = '''
SELECT COUNT(Mission_Outcome) AS SuccessOutcome
FROM SPACEXTABLE
WHERE Mission_Outcome LIKE 'Success%';
'''

pd.read_sql_query(task_7a, conn)
```

```
[55]:
```

SuccessOutcome	
0	100

```
[56]: %%sql

SELECT AVG(Payload_Mass__kg_)
FROM SPACEXTABLE;

* sqlite:///my_data1.db
Done.
```

```
[56]:
```

AVG(Payload_Mass__kg_)
6138.287128712871

```
[58]: task_7b = '''
SELECT COUNT(Mission_Outcome) AS FailureOutcome
FROM SPACEXTABLE
WHERE Mission_Outcome LIKE 'Failure%';
'''

pd.read_sql_query(task_7b, conn)
```

```
[58]:
```

FailureOutcome	
0	1

- **Key Findings:**
- A total of **100 successful mission outcomes** were recorded in the dataset.
- Only **1 mission outcome** was classified as a **failure**, indicating a very low mission failure rate.
- The results demonstrate the **high reliability and operational success** of SpaceX launch missions.

Boosters Carrying Maximum Payload

```
%%sql
```

```
SELECT Booster_Version, PAYLOAD_MASS_KG_  
FROM SPACEXTABLE  
WHERE PAYLOAD_MASS_KG_ = (  
    SELECT MAX(PAYLOAD_MASS_KG_)  
    FROM SPACEXTABLE  
);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

- **Multiple Falcon 9 Block 5 booster versions** carried the **maximum payload mass of 15,600 kg**.
- The results indicate that **Falcon 9 Block 5** consistently supported SpaceX's **highest payload capacity**.
- These boosters demonstrate the capability of handling **heavy payload missions** while maintaining operational reliability.

2015 Launch Records

```
%%sql
```

```
SELECT
    substr(Date, 6, 2) AS Month,
    substr(Date, 1, 4) AS Year,
    Landing_Outcome,
    Booster_Version,
    Launch_Site
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
    AND Date LIKE '2015-%';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Month	Year	Landing_Outcome	Booster_Version	Launch_Site
01	2015	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	2015	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- **Two drone ship landing failures** were recorded during **2015**.
- Both failed landings occurred at **CCAFS LC-40** using **Falcon 9 v1.1** booster versions.
- These missions provided valuable operational insights that contributed to improvements in subsequent landing attempts.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **"No attempt"** was the most frequent landing outcome with **10 occurrences** during the selected period.
- **Success (ground pad), Success (drone ship), and Failure (drone ship)** each occurred **5 times**.
- Less frequent outcomes included **Controlled (ocean), Uncontrolled (ocean), Precluded (drone ship), and Failure (parachute)**, showing the evolution of SpaceX landing strategies over time.

```
%%sql

SELECT Landing_Outcome,
       COUNT(Landing_Outcome) AS Total_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Total_Count DESC;
```

```
* sqlite:///my_data1.db
Done.
```

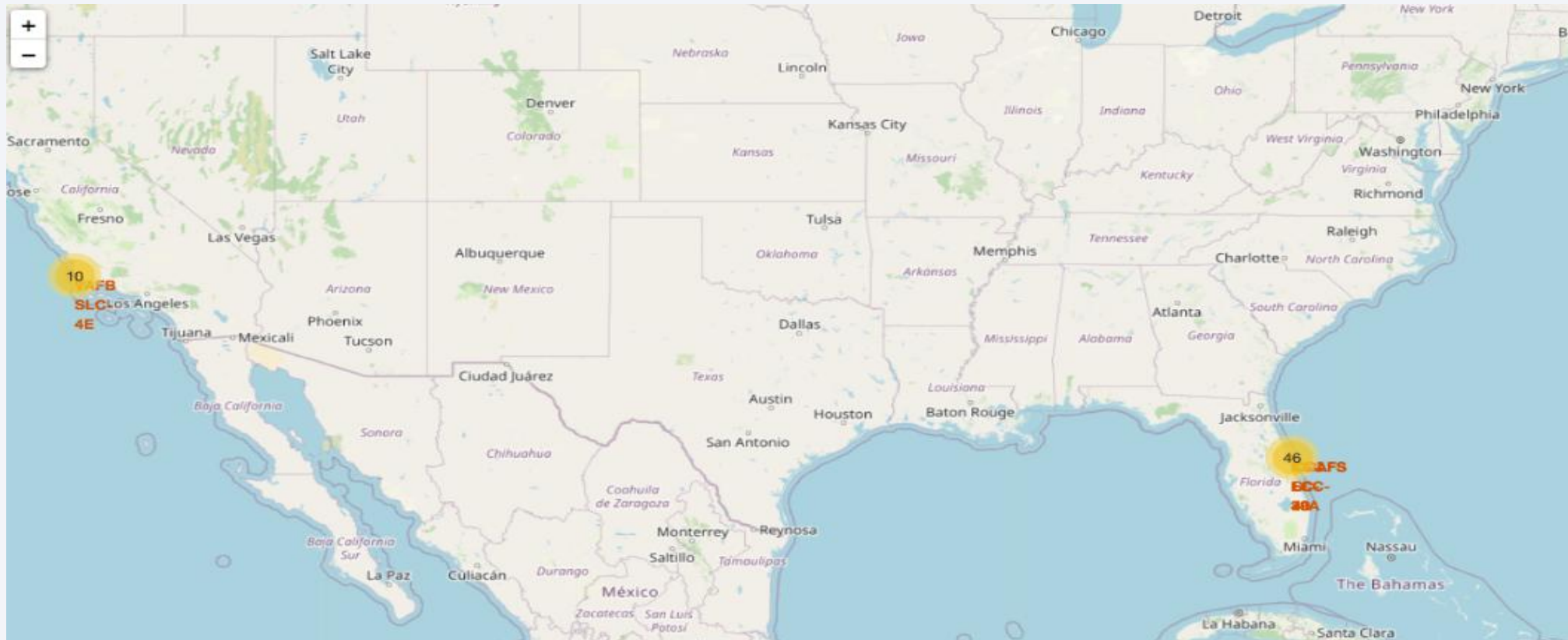
Landing_Outcome	Total_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and the glowing lights of cities and continents against the dark background of space. The Earth's surface is a mix of blue oceans and dark landmasses, with numerous bright yellow and white lights indicating urban areas and infrastructure.

Section 3

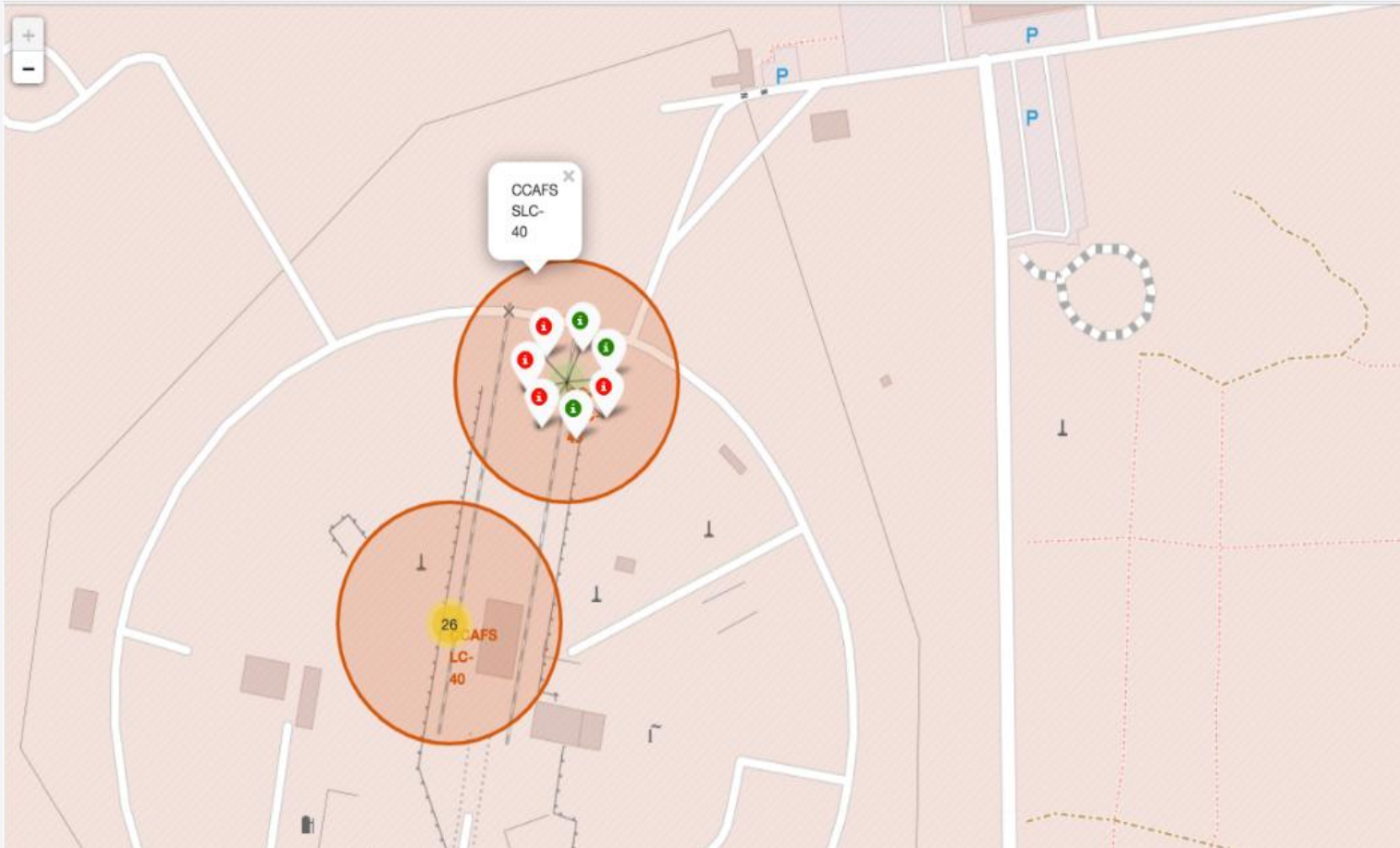
Launch Sites Proximities Analysis

Launch Site Locations



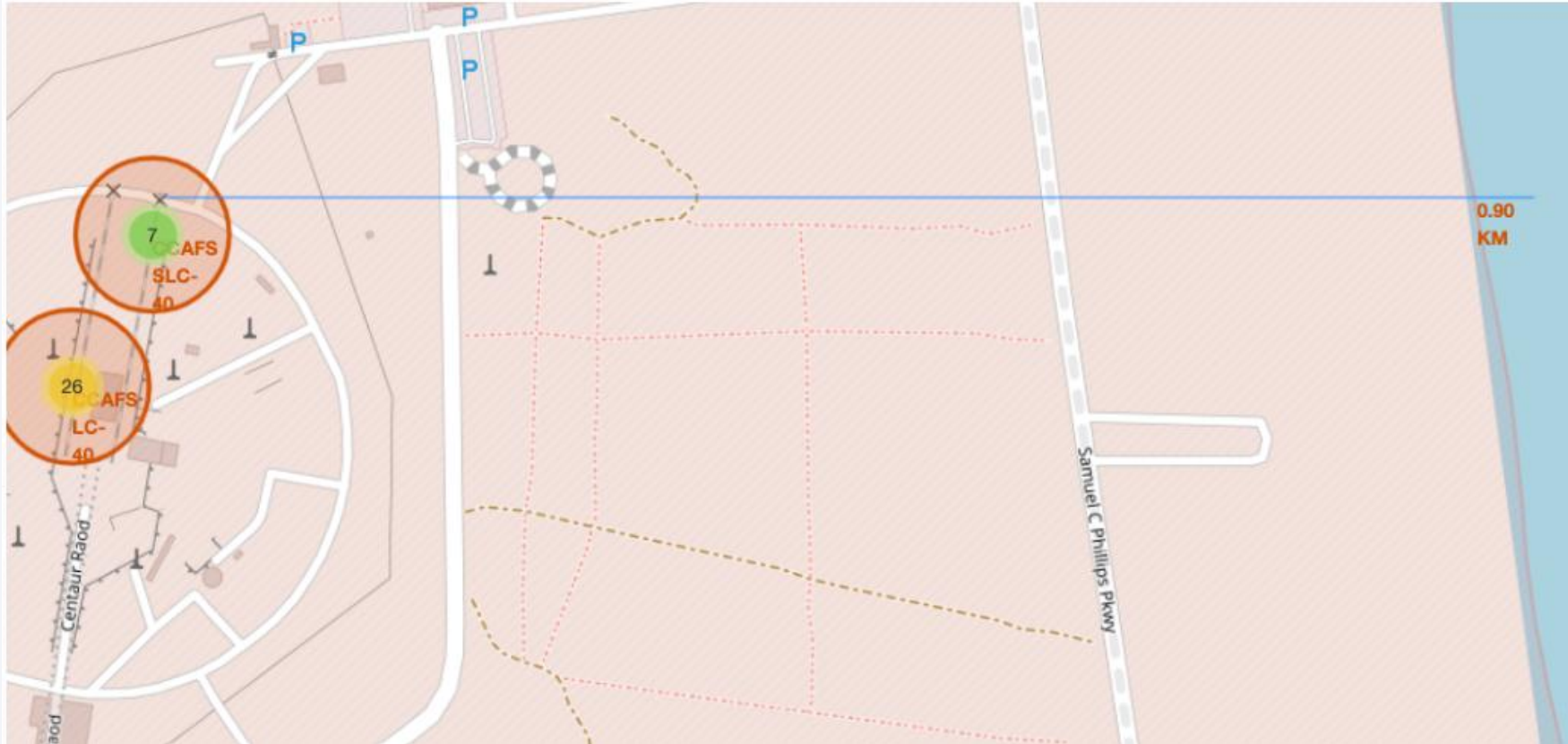
- All launch sites are located in **coastal areas** for safe rocket launches.
- **Vandenberg (California)** is the only West Coast launch site.
- **CCAFS** and **KSC (Florida)** host the majority of SpaceX launches.
- The Folium map provides an interactive visualization of launch site locations.

Interactive Marker Cluster Analysis

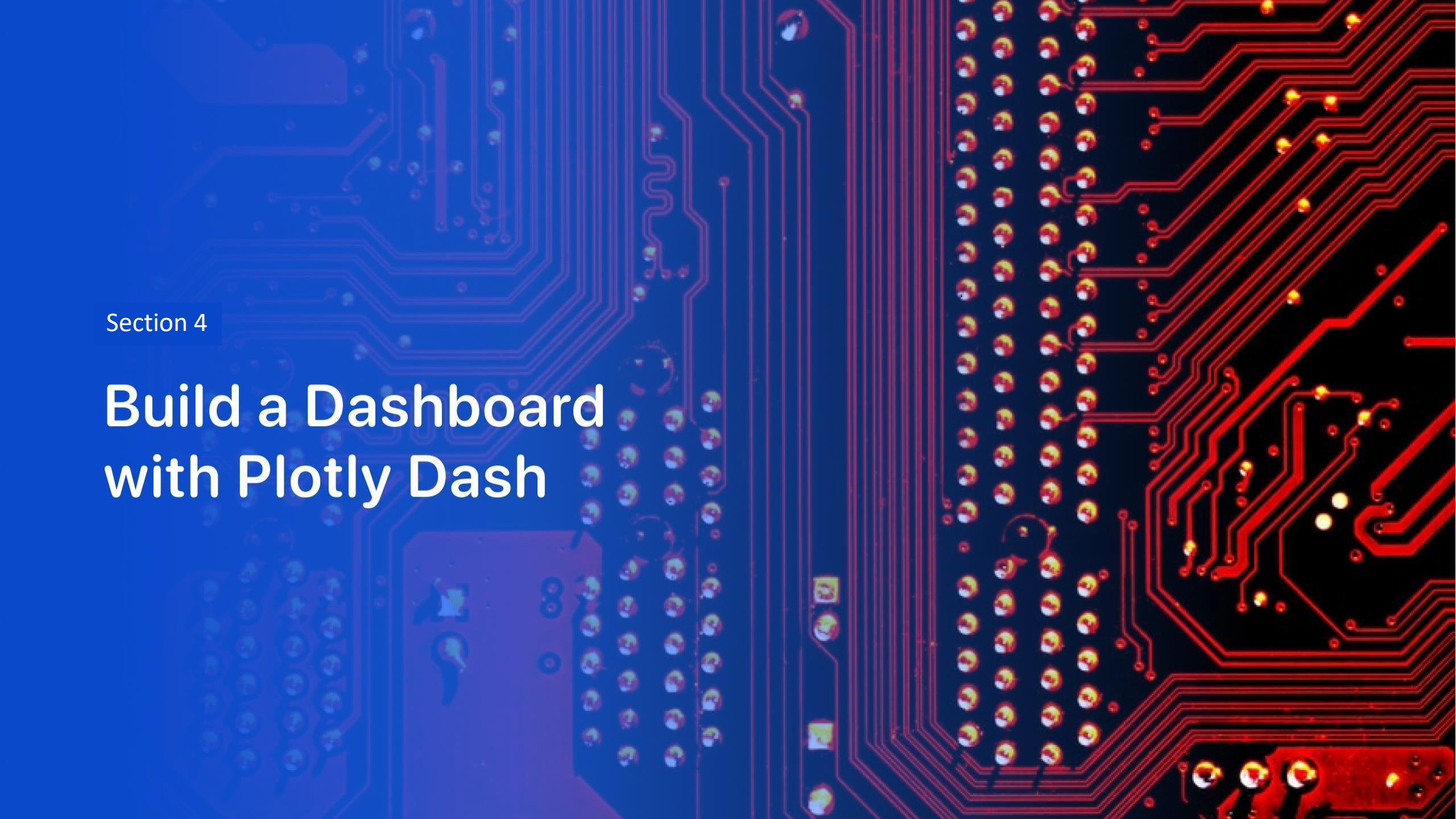


- Folium was used to visualize individual Falcon 9 launch records at **CCAFS SLC-40**.
- **Green markers** represent successful launches, while **red markers** represent failed launches.
- Marker clustering improves map readability when multiple launches occur at the same site.
- The visualization supports geographic analysis and helps compare launch outcomes across the launch site.

Distance of Launch Sites from Nearby Landmarks



- The proximity analysis shows that SpaceX launch sites are strategically located near the coastline, enabling safer launch operations and efficient mission planning.



Section 4

Build a Dashboard with Plotly Dash

Interactive Dashboard (Plotly Dash)

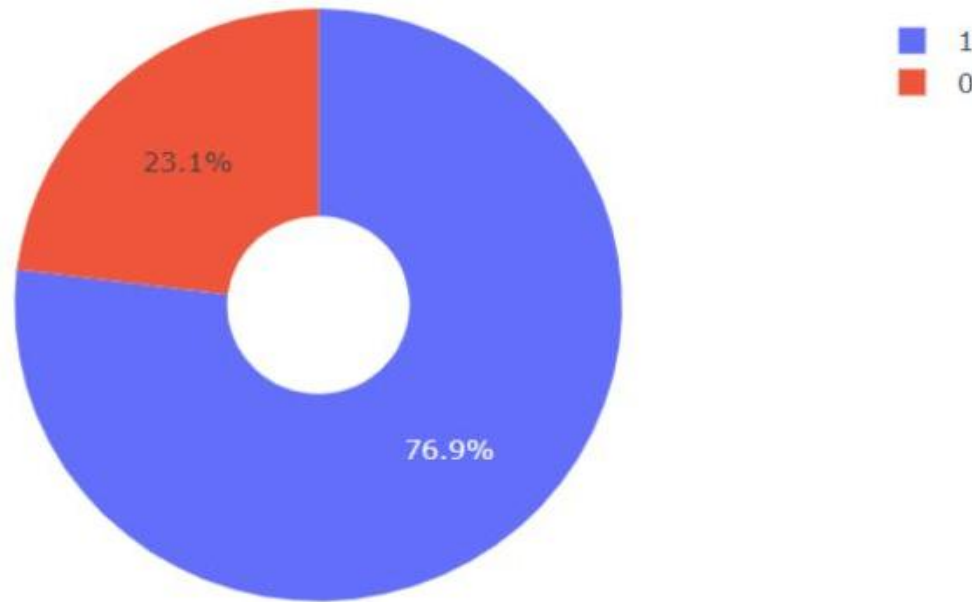
Total Success Launches By all sites



- Interactive dashboard developed using Plotly Dash.
- Displays successful launches across all launch sites.
- KSC LC-39A recorded the highest number of successful launches.
- Figure: Plotly Dash dashboard showing the success distribution across all launch sites.

Highest Landing Success Rate by Launch Site

KSC LC-39A: Success vs Failure

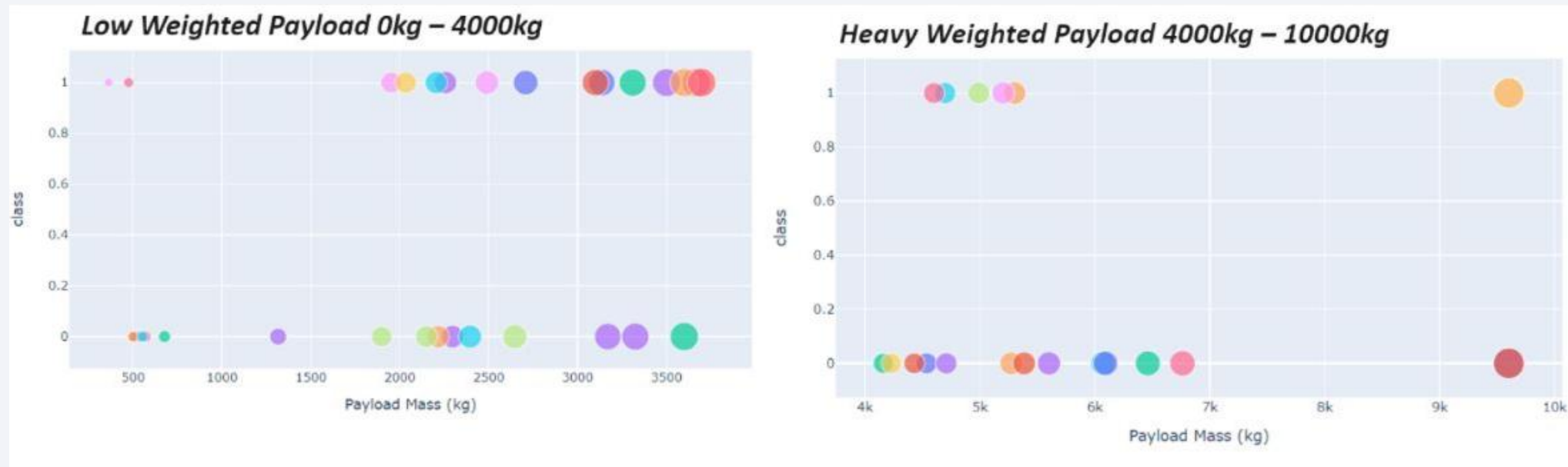


KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

- Shows the success and failure distribution for **KSC LC-39A**.
- Approximately **76.9%** of launches were successful.
- Helps compare mission outcomes at the selected launch site.
- Demonstrates the effectiveness of Falcon 9 operations at KSC LC-39A.

Payload Mass vs Launch Outcome

- FT booster versions achieved the highest landing success rate across the selected payload range.
- Most successful landings occurred for payloads between 2,000 and 4,000 kg.



Lower payloads generally achieved higher landing success rates than heavier payloads.



Section 5

Predictive Analysis (Classification)

Predictive Analysis (Classification)

- Key Finding: Decision Tree achieved the highest classification accuracy (88.89%) among all evaluated machine learning models, making it the most effective model for predicting Falcon 9 first-stage landing outcomes.

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV

parameters = {
    'criterion': ['gini', 'entropy'],
    'max_depth': [2, 4, 6, 8, 10],
    'max_features': ['sqrt', 'log2', None]
}

tree = DecisionTreeClassifier()

tree_cv = GridSearchCV(
    estimator=tree,
    param_grid=parameters,
    cv=10
)

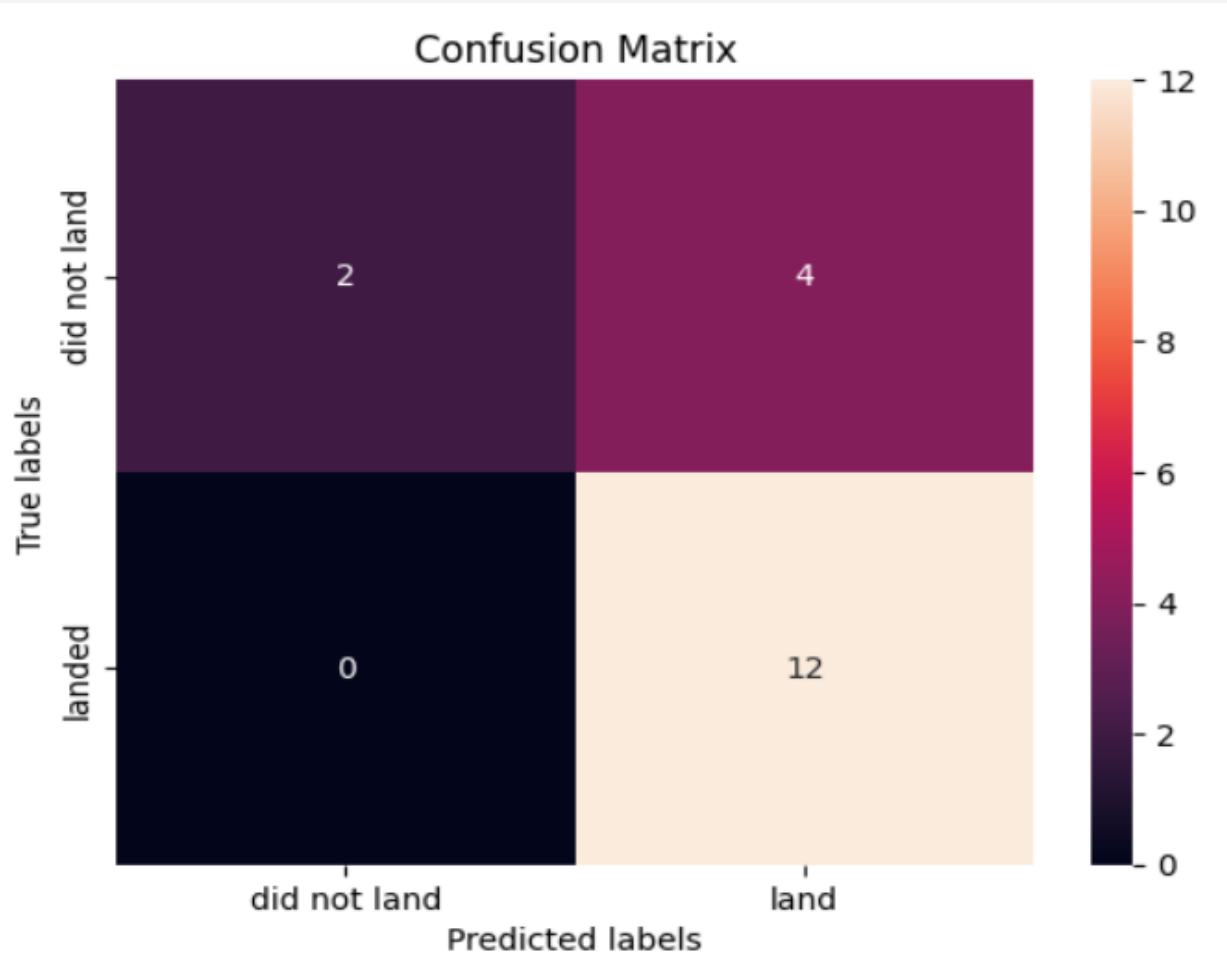
tree_cv.fit(X_train, Y_train)

print("tuned hyperparameters :(best parameters) ", tree_cv.best_params_)
print("accuracy :", tree_cv.best_score_)

print("Logistic Regression:", logreg_cv.score(X_test, Y_test))
print("SVM:", svm_cv.score(X_test, Y_test))
print("Decision Tree:", tree_cv.score(X_test, Y_test))
print("KNN:", knn_cv.score(X_test, Y_test))

Decision Tree = 0.8889 (88.89%)
```


Confusion Matrix



- Compared multiple classification models for predicting Falcon 9 landing success.
- **Decision Tree** achieved the highest accuracy among the evaluated models.
- The selected model was used for predicting future landing outcomes.

Conclusions

- SpaceX has achieved a high success rate in Falcon 9 first-stage landings, demonstrating significant improvements in reusable rocket technology.
- Launch success was influenced by factors such as **launch site, payload mass, booster version, and landing method**, with payloads between **2,000–4,000 kg** showing strong success rates.
- Geospatial analysis showed that launch sites are strategically located near coastal regions, supporting safe launch and recovery operations.
- Among the evaluated machine learning models, the **Decision Tree classifier achieved the highest prediction accuracy (88.89%)**, making it the best-performing model for predicting landing outcomes.
- Overall, the analysis demonstrates that **data analytics and machine learning** can effectively support the prediction and evaluation of Falcon 9 first-stage landing success.

Thank you!

